Activity 6 - Diagnose Hotspots and Set a Reduction Baseline

Alignment: K3 | A1 | LO3 Duration: 50 minutes Objective: Use inventory, lifecycle and business data to identify hotspots, set a defensible baseline and frame reduction opportunities.

Scenario

EcoForge's largest reported sources are purchased aluminium, product electricity use, imported electricity, air freight and refrigerant leakage. Data quality varies: electricity is metered, aluminium uses spend-based data, use-phase emissions rely on assumptions and freight distances are estimated.

Materials

- 2025 category inventory
- Product lifecycle facts
- Data-quality ratings
- Hotspot and baseline workbook

Questions

- Which sources are largest and which are most influenceable?
- Where can lifecycle burden shift occur?
- Which baseline metrics suit growth and efficiency decisions?
- What data improvements could change prioritisation?

Detailed procedure

1. Rank sources by absolute emissions, trend, data quality, organisational influence and stakeholder significance.
2. Map the product lifecycle from material extraction through manufacturing, transport, use and end of life.
3. Identify potential burden shifts, such as lightweight material that increases energy use or recycling claims that ignore yield loss.
4. Define an absolute baseline and at least one intensity metric with denominator rationale.
5. Separate near-term operational levers from supplier, product-design and customer-use levers.
6. Create an opportunity shortlist using abatement potential, feasibility, cost, lead time and evidence confidence.
7. Run a sensitivity check on the two weakest high-impact assumptions.
8. State which additional primary data is needed before investment approval.

Required evidence

- Hotspot dashboard
- Lifecycle map
- Baseline and intensity statement
- Prioritised opportunity shortlist

Acceptance check

Priorities reflect magnitude, influence and uncertainty; baseline and intensity denominator are explicit; no recommendation creates an undisclosed lifecycle burden shift.

Files

- Activity-06-Workbook.xlsx - editable working file
- Activity-06-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.